

UK PATENT APPLICATION

SYSTEM AND METHOD FOR AUTONOMOUS CROSS-DOMAIN CAUSAL DISCOVERY AND BUSINESS GENERATION USING ENSEMBLE PREDICTION AND TEST-DRIVEN DEVELOPMENT

FIGURES

The following simple flow diagrams describe the content of each figure for formal patent drawing conversion. All figures use standard patent drawing conventions: rectangular boxes for processes, diamond shapes for decisions, cylinders for databases, and directional arrows for data flow.

FIGURE 1: System Architecture Overview

Innovation (4) - Closed-Loop Autonomous Business Pipeline

Claims: 1, 11, 16

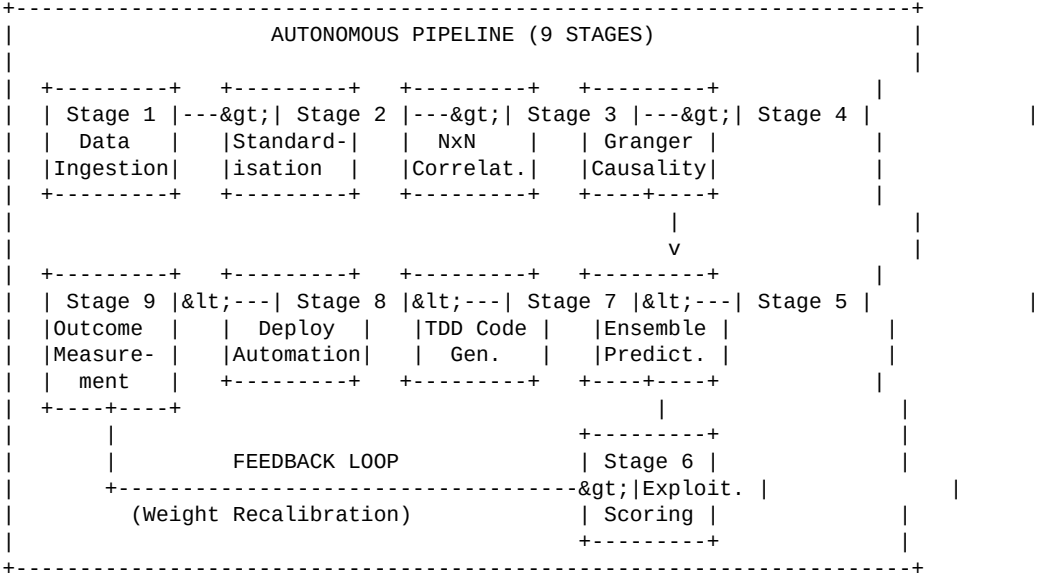


FIGURE 2: Multi-Source Data Ingestion Pipeline
Innovation (1) - Variable-Level Cross-Domain Causal Discovery
Claims: 1, 2, 11

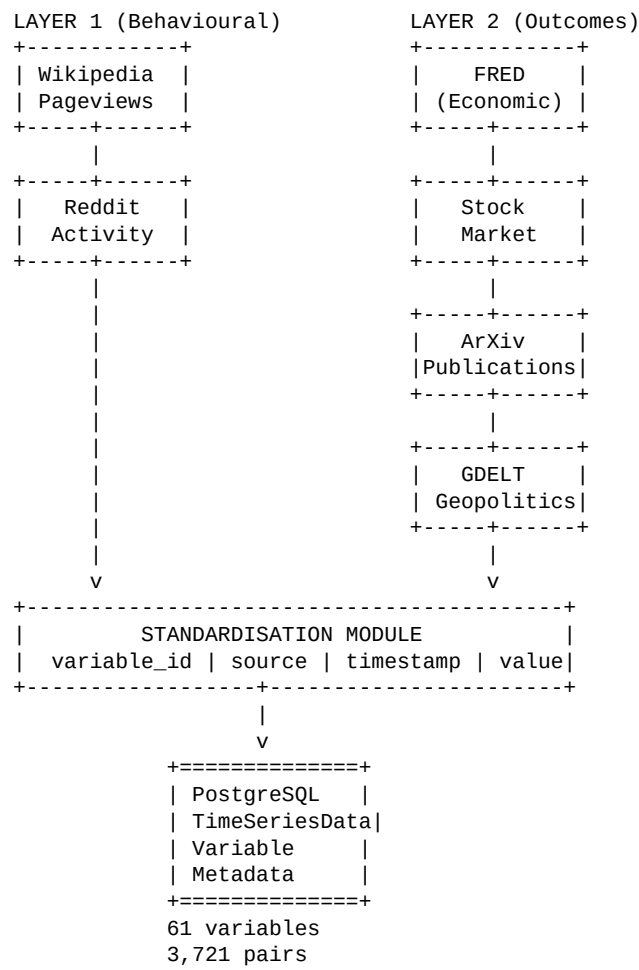


FIGURE 3: Cross-Domain Correlation Engine
Innovations (1) and (5)

Claims: 1, 3, 11

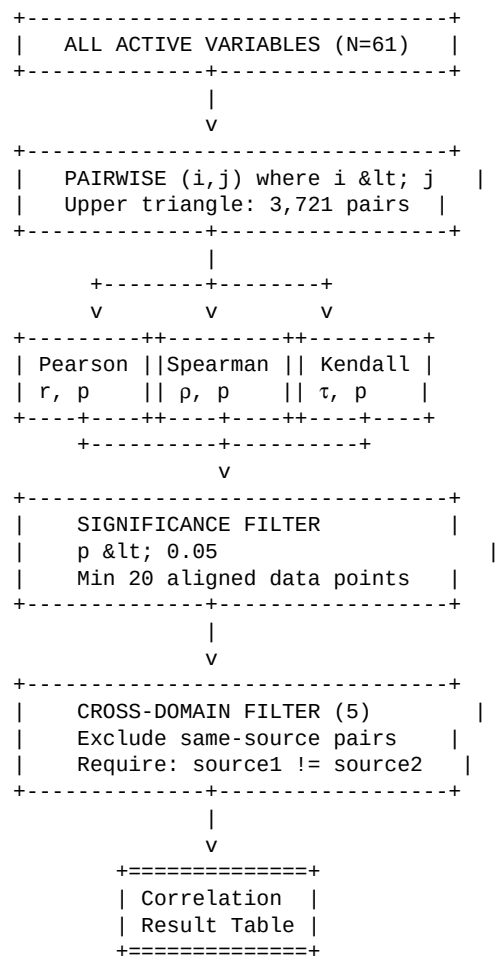


FIGURE 4: Bidirectional Granger Causality System

Innovation (1)

Claims: 1, 4, 11, 34

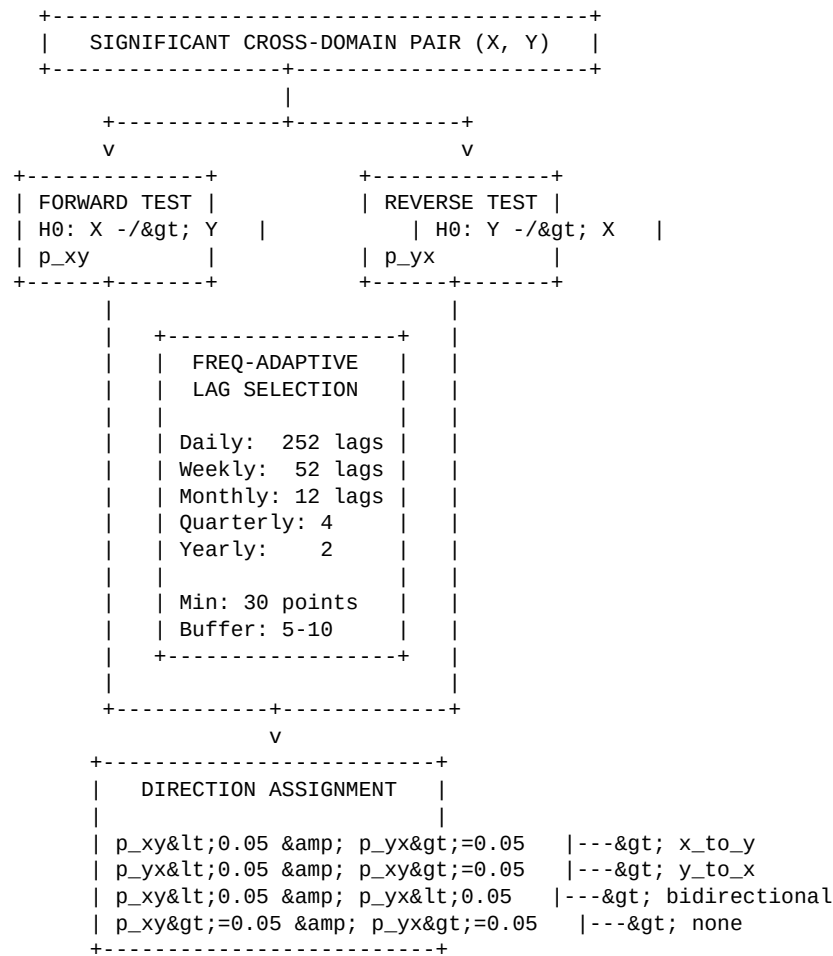


FIGURE 5: Three-Model Ensemble Architecture

Innovation (2)

Claims: 1, 5, 6, 11, 12, 21, 24

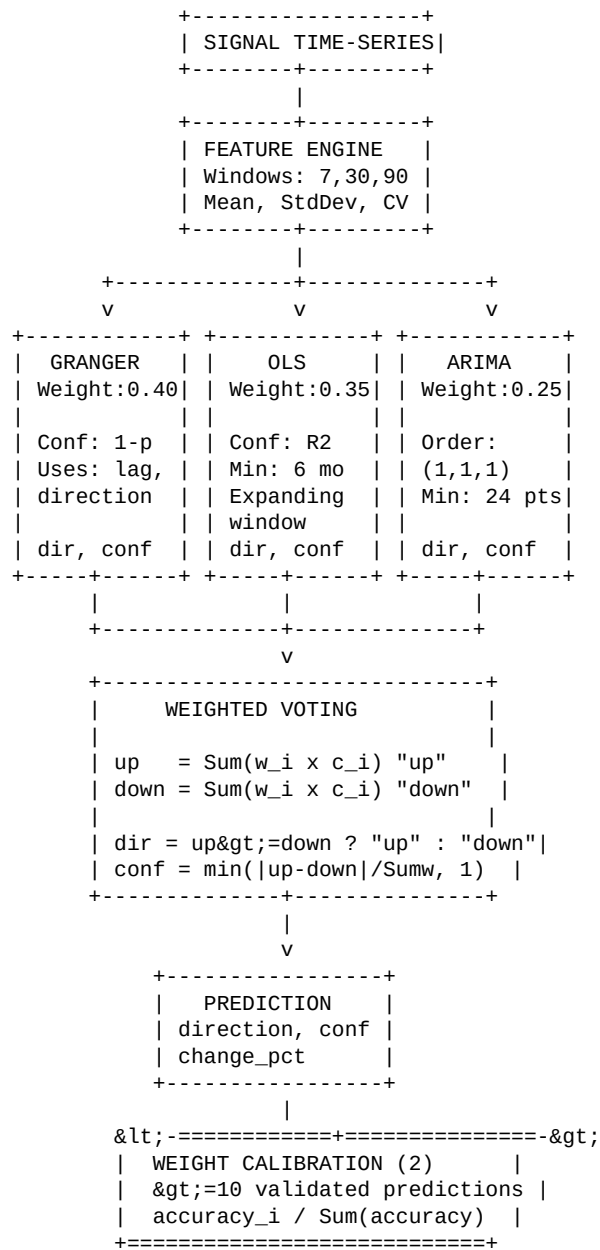


FIGURE 6: Cross-Layer Enforcement

Innovation (5)

Claims: 1, 7, 11, 20

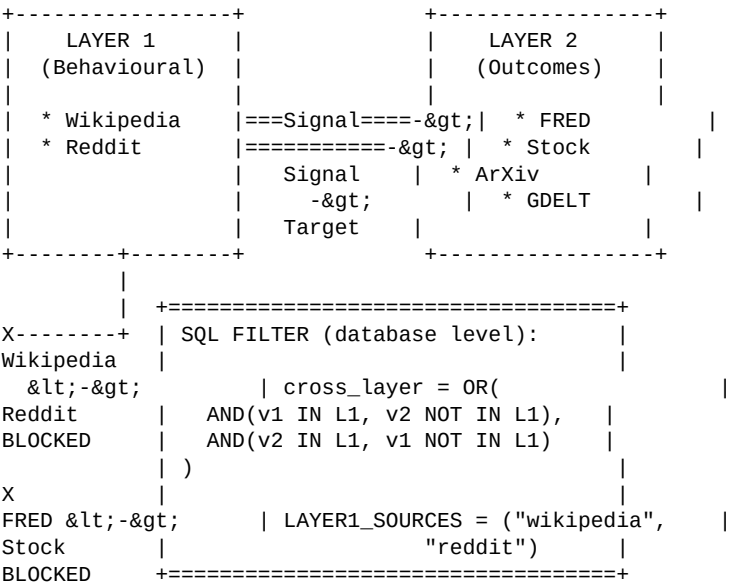


FIGURE 7: Exploitation Scoring Pipeline

Innovations (2) and (4)

Claims: 1, 8, 11, 17, 26

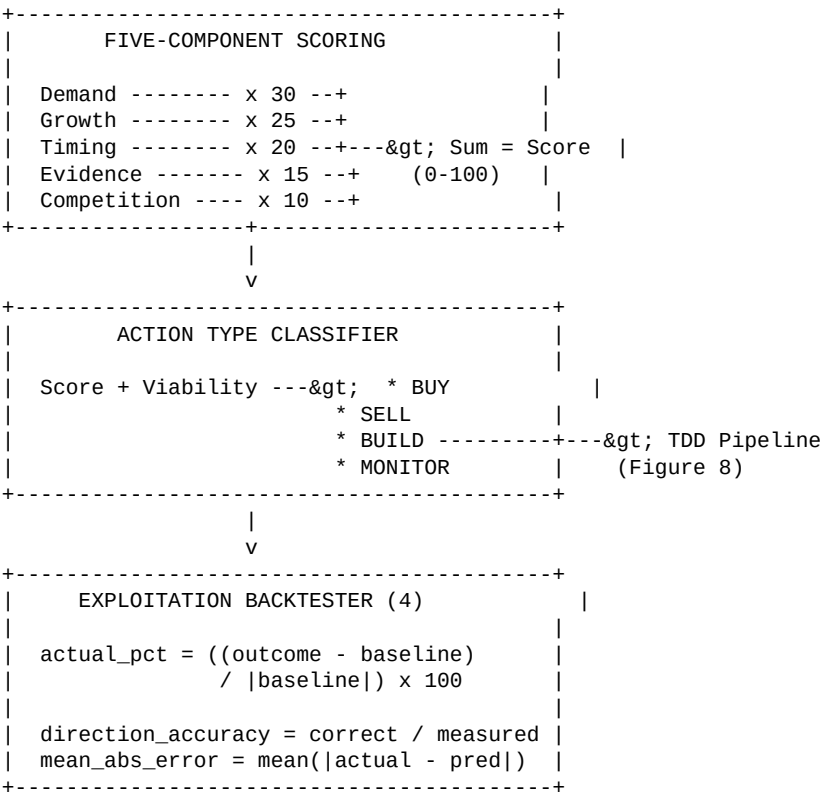


FIGURE 8: Autonomous TDD Code Generation Overview

Innovation (3)

Claims: 1, 9, 11, 13, 16, 28

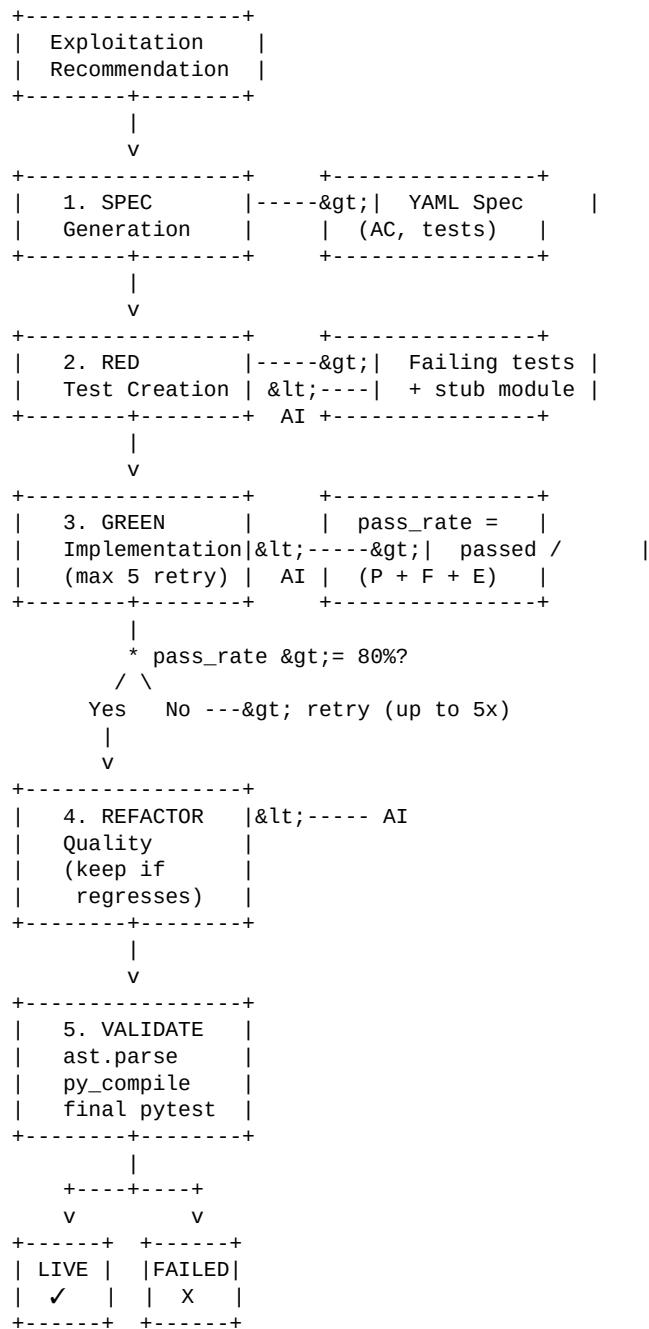


FIGURE 9: Five-Phase TDD Pipeline Detail

Innovation (3)

Claims: 9, 10, 13, 14, 22, 28

SPEC PHASE		RED PHASE	
+-----+		+-----+	
Keyword Analysis		Generate test files	
8 categories:		from YAML spec	
frontend, backend,			
database, auth,		Create stub module	
deployment, payments,		with NotImplemented-	
analytics, CRUD		Error for all imports	
Template Matching:		Verify: all tests	
Layer align: 30pts		fail (confirmed RED)	
Tag match: 15pts		+-----+	
Framework: 10pts		GREEN PHASE	
Desc words: 5pts		+-----+	
(max 20)		Extract symbols:	
		regex: from X import Y	
Complexity Tiers:			
LOW: 4 AC		Submit to AI:	
MEDIUM: 7 AC		• test code	
HIGH: 12 AC		• symbol list	
+-----+		• prev failure (if	
REFACTOR PHASE		retry attempt)	
+-----+			
AI improves quality		pass_rate >= 80%?	
		Yes -> REFACTOR	
Regression guard:		No -> retry (max 5)	
If new tests fail,		+-----+	
keep pre-refactor		VALIDATE PHASE	
version		+-----+	
+-----+		1. ast.parse (syntax)	
ERROR CATEGORIES		2. py_compile (all	
+-----+		files)	
* syntax		3. Final pytest run	
* test		4. Record metrics:	
* frontend		file_count,	
* import		line_count	
* wiring		5. Status: LIVE or	
* config		FAILED + error	
* runtime		breakdown JSON	
+-----+		+-----+	

FIGURE 10: Deployment Automation Flow

Innovations (3) and (4)

Claims: 1, 9, 11, 15, 16, 33

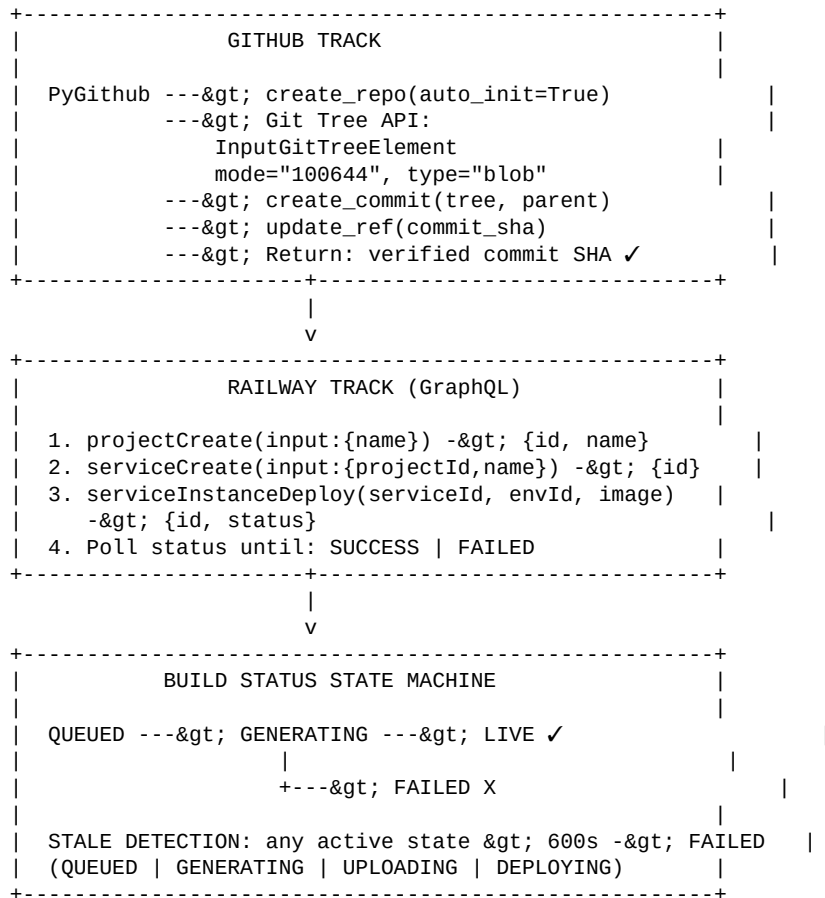


FIGURE 11: Outcome Measurement and Feedback Loop Innovations (2) and (4)

Claims: 1, 5, 6, 11, 12, 16, 17, 25

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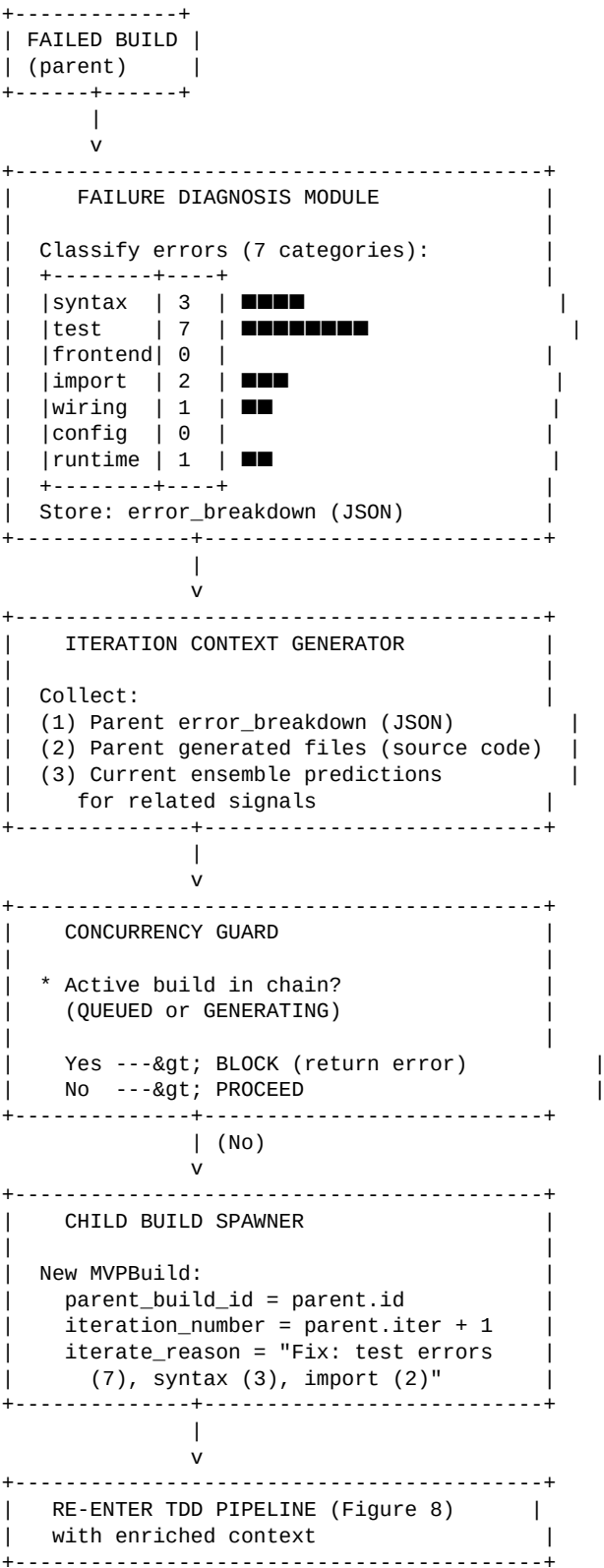
+-----+
| TRACK 1: PREDICTION VALIDATION (Daily 02:30 UTC) |
| Query: status='pending' AND target_date <= today |
| Fetch: actual value from TimeSeriesData |
| Compute: actual_direction, actual_change_pct |
| Set: direction_correct = (predicted == actual) |
| Update: status -> 'validated' |
+-----+
|
+-----+
| TRACK 2: WALK-FORWARD BACKTEST (Mon 04:00 UTC) |
| For split_idx = 6 to len(data)-1: |
|   train = data[0:split_idx] |
|   slope, intercept = polyfit(x_train, y_train, 1) |
|   predicted_y = slope x x_next + intercept |
|   direction_correct = (pred_dir == actual_dir) |
|   R2 = 1 - (ss_res / ss_tot) |
|   Store: model_version='backtest_walkforward' |
+-----+
|
+-----+
| TRACK 3: EXPLOITATION BACKTEST |
| Window: created_at -> created_at + (months x 30d) |
| actual_pct = ((outcome-baseline)/|baseline|) x 100 |
| Direction accuracy across all measured recs |
| Mean absolute error: mean(|actual - predicted|) |
+-----+
|
v
+-----+
| WEIGHT RECALIBRATION |
| Ensemble: accuracy_i / Sum(accuracy) -> new weights |
| Exploit.: discriminative components -> boost ±5pts |
| +-----+ |
| | Feeds back to Ensemble Engine (Figure 5) | |
| +-----+ |
+-----+

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FIGURE 12: Build-Iterate Loop

Innovation (6)

Claims: 28, 29, 30, 31



ITERATION CHAIN TIMELINE:

+-----+	+-----+	+-----+
v1	----> v2	----> v3
FAILED	FAILED	LIVE
+-----+	+-----+	+-----+

FIGURE 13: Autonomous Scheduling Timeline

Innovation (4)

Claims: 1, 11, 16, 23, 27

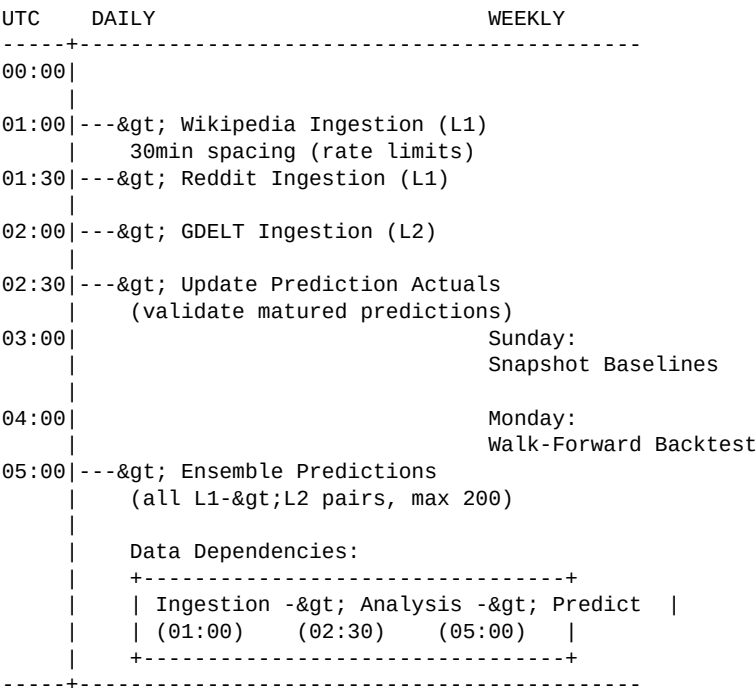
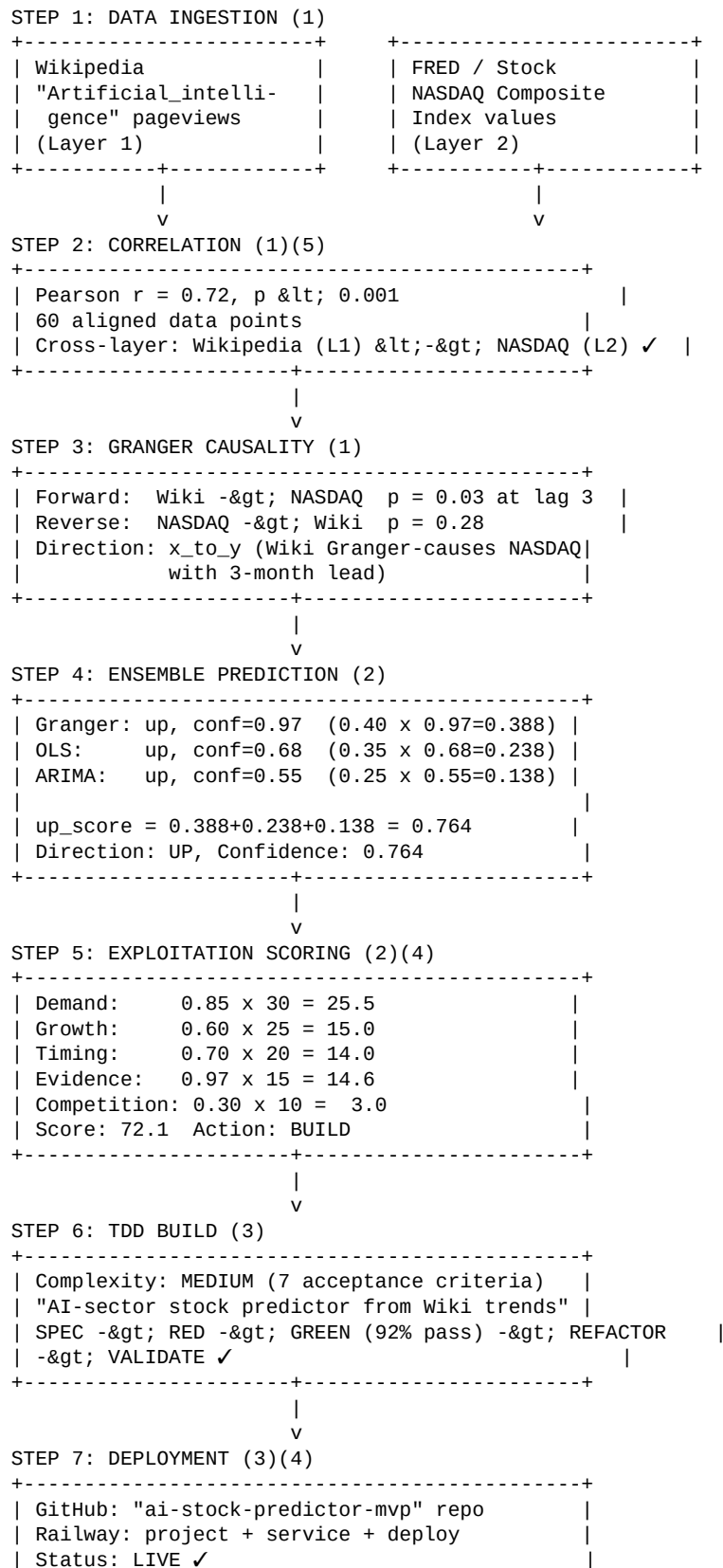


FIGURE 14: End-to-End Example Scenario

All Innovations: (1)(2)(3)(4)(5)(6)

Claims: 1, 11, 16



| URL: <https://ai-stock-predictor.up.railway.app>|

+-----+

|

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STEP 8: OUTCOME VALIDATION (2)(4)

+-----+

| +3 months (matching Granger lag 3): |

| Predicted: UP |

| Actual NASDAQ: +8.3% Direction: UP ✓ |

| direction_correct = True |

| Granger weight recalibrated: 0.40 -> 0.42 |

+-----+